Interlaboratory comparison of radiated emission measurements using a spherical dipole radiator

T.-W.Kang, Y.-C.Chung, S.-H.Won and H.-T.Kim

Abstract: An interlaboratory radiated emission comparison has been performed, comparing electric field strength measurements in the frequency range 30–1000MHz at a 10m open area test site. A spherical dipole radiator is employed as a reference emitter. Comparison results are presented in terms of measurement repeatability of electric field strengths measured at test sites for each frequency. Expanded uncertainty for the 95.45% confidence level is evaluated to be +5.6/–5.7dB and +4.9/–4.8dB for the frequency ranges corresponding to biconical and log-periodic antennas, respectively. Uncertainty analysis demonstrates that the uncertainty mainly stems from the antenna factor calibration, mismatch, the performance of a receiver and site imperfection.

1 Introduction

The issue of repeatability in radiated emission (RE) measurements has been addressed by many in the EMC community. Kolb performed a statistical comparison of site-to-site measurement reproducibility at 10m sites and presented measured results showing agreement within 3.8dB in the frequency range of 30–1000MHz [1]. A specially designed emitter capable of generating 101 frequencies spaced logarithmically was used for the work. Halaberda *et al.* compared RE test results collected from 44 sites including both open area test sites (OATSs) and semi-anechoic chambers for 3m and 10m distances [2]. In [2], it was demonstrated that OATS results have larger deviations from the mean than semi-anechoic chamber results and that over 60% of the sites had one or more frequencies with differences of over 5dB from the mean. A commercially available emission source with a repeatability within 0.5dB was used for the measurements in the 30–500MHz frequency range at increments of 10MHz [2]. Note that the two research groups dealt with the subject only for vertical polarisation.

To ensure consistency in RE compliance tests, the repeatability of measurement is of importance. This is especially valuable when the test of the same product produces different results at two different sites. A test laboratory should ensure the quality of results by monitoring test or calibration results. This monitoring should be planned and reviewed and may include participation in interlaboratory comparison or proficiency testing programmes [3].

The interlaboratory comparison presented in this paper aims to determine how reproducible RE measurements are under the constant radiated field condition. The result of the interlaboratory comparison of RE measurements using the spherical dipole radiator (SDR) as a reference emitter is presented and uncertainty analysis is carried out to take into account all possible sources of errors. All uncertainty values in this paper are stated at the 95.45% confidence level.

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2 System description

In the interlaboratory comparison of RE measurements, it is very important to use a reference emitter with high stability and good repeatability. In this Section, the characteristics of the SDR employed in the interlaboratory comparison and the setup of measurement system are described.

2.1 Spherical dipole radiator

A commercially available SDR (SRS2100, Seth Corporation, USA) with a repeatability of 0.2–0.5dB was used as a reference emitter. Its frequency range was 10MHz to 2.5GHz. The SDR system consists of a spherical radiator with a 10cm outer diameter and a control unit to interface to the operator and an external RF signal source. The radiator, i.e. a transmitting antenna whose radiation pattern is explicitly known, is electrically isolated from the RF source by optical-fibre cables. The use of the optical-fibre cables between the control unit and the radiator eliminates disruption of the radiated field which is to be measured. Details of the SDR can be found in [4].

Before the comparison test commenced, the RF input power for each frequency was determined in advance at the pilot site so that the subsequent gap voltage of the SDR was in a typical useful range of 0.2–1.0V. Moreover, during the procedure of determining the gap voltage, care had to be taken not to saturate the RF amplifier installed in the spherical radiator due to high RF input power. The gap voltage determined above is referred to as a reference gap voltage, shown in Fig. 1. The reference gap voltages for horizontal and vertical polarisations are nearly identical except at the frequencies of 150 and 1000MHz. To maintain the radiated power as constant as possible at each site,

the gap voltage of the SDR has been monitored within +12/–5mV and +36/–22mV with respect to the reference gap voltage for horizontal and vertical polarisations, respectively, as shown in Fig. 1.

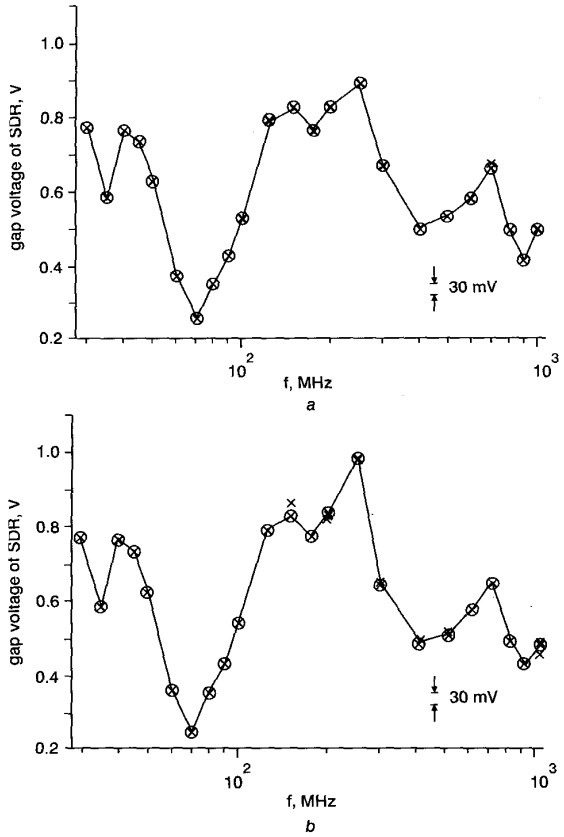


Fig. 1 *Gap voltages of the spherical dipole radiator*
—○— reference gap voltage
× gap voltages monitored at the 12 OATSs
a Horizontal polarisation
b Vertical polarisation

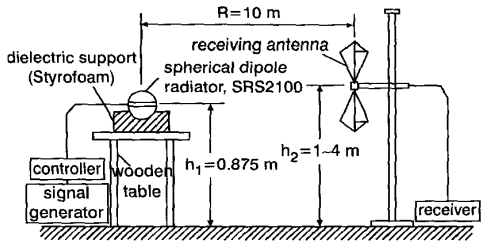


Fig. 2 *Setup of radiated emission measurements*

2.2 Measurement setup

A typical measurement setup is illustrated in Fig. 2. The SDR is positioned at the centre of a turntable on a 0.8m-high wooden table [5]. Because of the round shape of the radiator, a dielectric support made of high-density Styrofoam was used to fix the spherical radiator on the flat table. To confirm that the support did not affect the radiated field to be measured, scattering parameters of the support were measured using a network analyser (HP8753C): $|S_{11}| \leq -47.5$dB, $|S_{21}| \leq -0.004$.dB in the frequency range 30–1000MHz.

The wooden table also affected the electric fields to be measured, as had been investigated extensively using an experimental approach [6]. Consideration of the effect of the table on uncertainty makes the associated analysis complicated. Therefore, it is assumed that the effect of the dielectric table is implicitly incorporated into the measured, as electric field and subsequently the influence is accounted for by measurement repeatability, as discussed later. The total height of the radiator above the metal ground plane is 0.875m.

Measurements were made at distance 10m for both horizontal and vertical polarisations. For vertical measurements, the SDR was aligned on the wooden table so that its equator, a band made of Teflon was parallel to the ground plane, whereas for horizontal measurements the radiator was rotated by 90° where the rotation axis coincides with the measurement axis defined as a horizontally interconnecting line between the centre of the SDR and the receiving antenna. The receiving antenna was scanned from 1 to 4m and maximum field strengths were recorded [5]. The receiving antenna could be a set of biconical and log-periodic antennas, or a bilog antenna to cover the full frequency range of interest. An electromagnetic interference (EMI) receiver capable of measuring the electric field strengths was used. EMI receivers manufactured by Rohde & Schwarz GmbH & Co. were used at all test laboratories. The receiver was used in the quasi-peak mode with 120kHz bandwidth.

The antenna factor (AF) of the receiving antenna should be added to the reading of the EMI receiver to convert it into electric field strength. Although the AF is dependent on the test environment, and especially on height, the free-space AF is used and the variation of AF with height is considered as a contribution to measurement uncertainty. The losses in the RF cables used for instrumentation also should be compensated.

In the comparison test, the SDR system was carried site to site to make on-site measurements. At each site, the measurement system was configured in accordance with international EMI regulations [5, 7]. Since the spherical dipole antenna with a uniform excitation voltage at the equator radiates as a dipole antenna, rotation of the turntable was not necessary [4]. Because of the limited duration of the battery of the SDR, the measurements for the selected frequencies were made for approximately two hours at each site. 11 test laboratories participated in the interlaboratory comparison and the number of sites was 12.

3 Comparison results

Measurements have been made for 23 discrete frequencies selected from the frequency list of ANSI standard C63.4 [7]. At several test sites, one or two test frequencies were skipped because of high external ambient noise. It is preferable to compare the measurement results with an absolute standard: the absolute standard does not exist since there is no ideally perfect OATS. Therefore, the mean of measured electric field strengths has to be the best alternative.

To define repeatability in the comparison test, the electric-field strengths measured at each site are considered as those obtained from repeated measurements. The standard uncertainty, or repeatability, for repeated measurements has been defined as follows. With n independent single measurements, the estimated value x_i of the true value of X_i is given by the arithmetic mean of the individual measured values $v_{i,j}$ ($j = 1, ..., n$)

$$x_i = \bar{v}_i = \frac{1}{n}\sum_{j=1}^{n} v_{i,j} \qquad (1)$$

where $v_{i,j}$ denotes the field strength of ith frequency at jth

test site. The estimated standard deviation s_{v_i} of the distribution of the measured values v_i is

$$s_{v_i} = u(v_i) = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(v_{i,j} - \bar{v}_i)^2} \qquad (2)$$

When repeated measurements are made, the standard deviation of the mean s_{x_i} is

$$s_{x_i} = u(x_i) = \frac{s_{v_i}}{\sqrt{n}} = \sqrt{\frac{1}{n(n-1)}\sum_{j=1}^{n}(v_{i,j} - \bar{v}_i)^2} \qquad (3)$$

Here, s_{x_i} is referred to as repeatability of RE measurements at the ith measurement frequency. Eqn. 3 is exactly true only if n is infinite. In practice, most standard laboratories either use $n \geq 10$ or apply the appropriate multiplying factors given in Table 1. In the present case, n is 5 or 10–12 and the multiplying factor is applied to s_{x_i} to obtain repeatability if $n < 10$.

Table 1: Multiplying factors

Number of measurements	Multiplying factor	Number of measurements	Multiplying factor
2	6.99	6	1.33
3	2.27	7	1.26
4	1.66	8	1.22
5	1.44	9	1.19

It should be emphasised that the gap voltage variation contributes not to the RE-measurement uncertainty of a specific laboratory but to the measurement repeatability in the interlaboratory comparison. The repeatability of the SDR given by the manufacturer must be considered to evaluate the uncertainty in RE measurements of a laboratory itself (see Table 2, transmitting part). On the other hand, the gap voltage variation shown in Fig. 1 influences the repeatability of the interlaboratory comparison. Moreover, note that the gap-voltage variation has already been incorporated into the measurement repeatability described above. The reason is that the gap voltage variation affects linearly the electric field to be measured because the reference gap voltage has been determined in the linear operating range of the SDR. For extremely large variations of the gap voltage, the interlaboratory comparison becomes meaningless.

Fig. 3 illustrates the measurement results for horizontal polarisation. The mean of the measured field strengths varies from 29.7 to 64.6dBμV/m. Note that the number of measured field strengths is 10–12 at all frequencies except at 100MHz. As mentioned above, because of ambient noise the number of measured field strengths at 100MHz is five, which is considerably less than that for other frequencies. The maximum deviation of measured field strengths from the mean is +5.7/–5.3dB. The broken line near the frequency of 100MHz observed in the graph of deviations, indicates that the test frequency between the two lines has been skipped due to ambient noise. As observed in Fig. 3, the deviation is relatively large at low frequencies; this large deviation results from the high voltage standing wave ratio (VSWR) of the receiving antenna at the corresponding frequency.

Fig. 4 shows the measured field strengths and their deviations from the mean for vertical polarisation. Comparing

Table 2: Uncertainty in RE measurements at the distance of 10m for a specific OATS among the 12 sites

Contributions to measurement uncertainty	Probability distribution function	Tolerance (dB) for a biconical antenna	Tolerance (dB) for log- periodic antenna
Transmitting part			
(*a*) Absolute level of accuracy of a signal source	rectangular	0.5	0.5
(*b*) Repeatability of the SDR	rectangular	0.5	0.5
(*c*) Mismatch between a signal source and the SDR: ($\lvert\Gamma_{SG}\rvert = \lvert\Gamma_{SDR}\rvert = 0.3333$)	U-shaped	+0.9/–1.0	+0.9/–1.0
(*d*) Measurement-distance variation	rectangular	0.4	0.4
Receiving part			
(*a*) Antenna-factor (AF) calibration	normal ($k = 2$)	2.5	2.5
(*b*) Cable-loss measurement	normal ($k = 2$)	0.3	0.3
(*c*) AF variation with height	rectangular	1.5	0.5
(*d*) AF interpolation against frequency	normal ($k = 2$)	0.3	0.3
(*e*) Antenna directivity	rectangular	+0.2/–0.0	+0.5/–0.0
(*f*) Antenna-phase-centre variation	rectangular	0.0	0.2
(*g*) Mismatch between a receiving antenna and a receiver: ($\lvert\Gamma_{Rcvr}\rvert = 0.3333$)	U-shaped	+1.7/–2.1 ($\lvert\Gamma_{Ant}\rvert = 0.6429$)	+0.9/–1.0 ($\lvert\Gamma_{Ant}\rvert = 0.3333$)
(*h*) Receiver specification:			
voltage accuracy	rectangular	1.0	1.0
absolute pulse response	rectangular	0.0	0.0
pulse-repetition response	rectangular	0.0	0.0
Open-area test site			
(*a*) Site imperfection	triangular	+3.2/–3.0	+3.2/–3.0
(*b*) Overall system repeatability	standard deviation	0.5	0.5
Combined standard uncertainty (dB)	normal	+2.6/–2.7	+2.3/–2.2
Expanded uncertainty (dB), $k = 2$		+5.2/–5.4	+4.5/–4.4

the results with those for horizontal polarisation shown in Fig. 3, the variation of mean against frequency is relative small, 43.9–64.0dBμV/m. The maximum deviation from the mean is +4.7/–5.7dB. In Figs. 3 and 4, dotted lines indicate the upper and lower bounds of measurement uncertainty, discussed later.

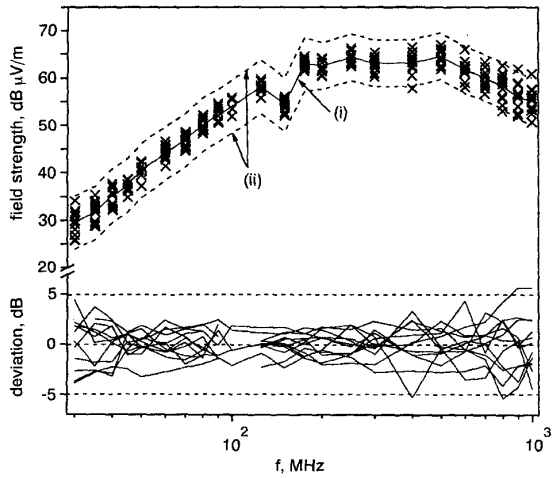


Fig. 3 *Measurement results for horizontal polarisation*
(i) mean
(ii) expanded uncertainty for 95.45% confidence level

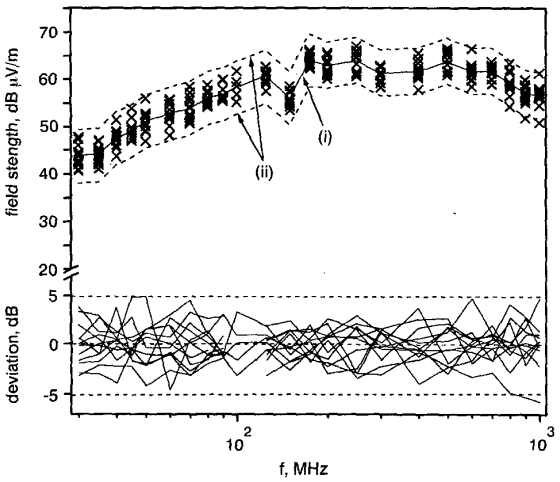


Fig. 4 *Measurement results for vertical polarisation*
(i) mean
(ii) expanded uncertainty for 95.45% confidence level

Comparing the behaviour against frequency of the measured electric field as shown in Figs. 3 and 4 it is seen that, the horizontal electric field is relatively lower than the vertical one at low frequencies. The exact theoretical behaviour of horizontal and vertical electric dipoles above an ideal OATS under the condition of constant radiated power has been calculated previously [8] and the results from this experimental work show good agreement. Here, note that the gap voltages, i.e. radiated RF power, of the SDR for both polarisations are almost same as that presented in Figs. 1 and 2. However, since the sensitivity of the gap voltage of the SDR to the RF input power depends on the frequency, the resultant radiated RF power varies with the frequency. This is why the curves of the mean of measured electric field strengths for horizontal and vertical polarisations do not vary smoothly with frequency.

Fig. 5 summarises the repeatability of RE-comparison tests obtained by using eqn. 3 and the multiplying factors shown in Table 1. Since the number of measured field strengths at 100MHz is five, which is rather small compared with the number for other frequencies, the repeatability at the corresponding frequency is not so good. Nevertheless, the repeatability is 2.1 and 3.2dB (2σ) for horizontal and vertical polarisations, respectively, at the worst-case frequency of 100MHz. Considering the uncertainty of 5.7dB in RE measurements discussed in Section 4, repeatability of the comparison test can be regarded as good for both polarisations.

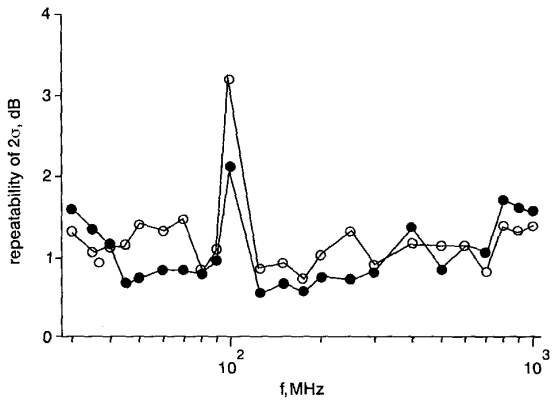


Fig. 5 *Repeatability of RE comparison tests*
—●— horizontal polarisation
—○— vertical polarisation

4 Measurement uncertainty

The uncertainty has been evaluated in accordance with the ISO guide for the expression of uncertainty in measurement [9]. Since the instrumentation is different from site to site, so is the associated measurement uncertainty of each site, though not noticeably. The individual contributions to uncertainty are dealt with in the order shown in Table 2 in which uncertainty in RE measurements at the distance of 10m for a specific OATS among the 12 sites is summarised. Since the numbers in the last two columns in Table 2 are not divided by the weighting factor for the specified probability distribution function, the corresponding numbers are referred to as tolerance. Note that the weighting factors are $\sqrt{3}$ for rectangular, $\sqrt{2}$ for U-shaped, $\sqrt{6}$ for triangular, 2 for normal, and 1 for standard deviation.

To evaluate uncertainty in RE measurements, contributions to uncertainty are grouped into three parts: transmitting and receiving parts, and the OATS itself. For the sake of completeness, each contribution is described briefly and details of each contribution are found in [10–12].

4.1 Transmitting part

The absolute accuracy of the RF signal generator and the repeatability of the SDR are obtained from the manufacturer's specification. The next contribution to uncertainty is the mismatch between the SDR and the RF signal generator. The mismatch uncertainty in the power transfer between a generator and a load is

$$u_{mis} = \frac{20\log(1 \pm |\Gamma_G||\Gamma_L|)}{\sqrt{2}} \quad (4)$$

where Γ_G and Γ_L are the reflection coefficients of the signal generator and the load, respectively, and the U-shaped distribution is assumed. To obtain the mismatch uncertainty

shown in Table 2, the value of voltage standing wave ratio (2:1) is used for the signal generator and the SDR.

The uncertainty for distance variation is the error in the absolute measurement of the distance between the SDR and the receiving antenna. During the measurement, the distance was measured with the tolerance of 5.0cm. Since the measurement distance is 10m, the receiving antenna is placed in the far-field region where the inverse-distance law is valid. Therefore, the resulting tolerance of the electric-field strength is 0.04dB.

4.2 Receiving part

The uncertainty of AF calibration is 2.5dB and that of cable loss measurement is 0.3dB where the coverage factor is 2. The phase centre of a log-periodic antenna varies with the frequency, which causes uncertainty in RE measurements. Since different types of antenna have been used at each site, the uncertainty due to the phase centre variation is 0.2–0.3dB.

The AF of an antenna varies with the antenna height because of mutual coupling with the ground plane. This effect on RE measurements is not specified by manufacturers. The numeric values given in Table 2 are attainable [12]. Because of smooth variation of the AF with the frequency for commercially available antennas, it is obvious that the uncertainty due to AF interpolation with frequency is small, as shown in Table 2. The angle of incidence with respect to the antenna boresight decreases as the measurement distance increases. Since the antenna calibration is carried out for the direction of boresight, it is reasonable to assume that the actual field strength could be stronger than the receiver reading by an unknown amount [10]. Although this influence is greater for vertical polarisation than for horizontal polarisation, the contribution of the antenna directivity to uncertainty is set to be 0.2 and 0.5dB for a biconical and a log-periodic antennas, respectively, irrespective of the polarisation [12].

The uncertainty due to mismatch between the receiver and the receiving antenna is obtained by eqn. 4. In general, the receiving antenna is composed of biconical and log-periodic antennas. The associated transition frequency between the two antennas is either 200 or 300MHz and the field strength measured by the log-periodic antenna is chosen at the transition frequency. A bilog antenna with a frequency range of 30–1000MHz was also used at several test sites. The VSWR of the bilog antenna deteriorates as the frequency reduces. If the maximum VSWR is chosen in the full operating frequency range of the bilog antenna, the uncertainty due to mismatch in its relatively high frequency range will be overestimated unnecessarily. For such cases, the transition frequency is arbitrarily set to be 300MHz to evaluate the uncertainty. The typical VSWR of the receiver under the condition of 0dB RF attenuation is 2:1 while those of biconical and log-periodic antennas are 4.6:1 and 2.0:1, respectively.

The uncertainty of the receiver is composed of voltage accuracy, absolute pulse response and pulse repetition rate response. The accuracy of measurement of sine-wave voltages is better than 1–1.5dB when supplied with a sine-wave signal at a 50Ω resistive source impedance. Since the transmitting source produces a continuous-wave (CW) signal, neither absolute pulse response nor pulse repetition response contribute to the combined uncertainty. However, for actual RE tests, an appropriate tolerance for the pulse response should be assigned.

4.3 Open area test site

The uncertainty due to OATS imperfection is generally unknown. The deviation of normalised site attenuation from the theoretical values gives an indication of the potential effect of site imperfection. However, it cannot be applied as a correction to the measured data.

The RE measurements have been performed at ten 10m OATSs and at two 30m OATSs. The sizes of the ground planes of the 10m OATSs were (18–23m) (L) × (8–18m) (W). For one of the two 30m OATSs, the ground plane of the 10m distance measurement was 22m (L) × 18m (W) and for the 30m distance measurement a narrower wire mesh of dimensions 38m (L) × 8m (W) was extended along the measurement axis. The other 30m OATS had the ground plane of 60m (L) × 52m (W). The ground planes of all sites consisted of metal plates and galvanised wire mesh with a spacing of less than 5.0mm. The flatness of all sites was satisfied by the Rayleigh criterion in [13]. The deviation of normalised site attenuation from the theoretical values for the 12 sites was within +1.7~+3.7/–3.3~–2.2dB and the associated distribution function is assumed to be triangular. The overall repeatability of the measurement system is typically 0.5dB.

Table 2 summarises the uncertainty in RE measurements at the distance of 10m for a particular OATS among the 12 OATSs. The uncertainty analysis demonstrates that uncertainty in RE measurements mainly originates from the AF calibration, mismatch, the performance of the receiver and site imperfection.

The evaluated uncertainty for each site depends on its instrumentation and the resultant expanded uncertainty is within +4.7~+5.6/–5.7~–5.0dB and +4.2~+4.9/–4.8~–4.2dB for the frequency ranges of biconical and log-periodic antennas, respectively. The maximum expanded uncertainty of +5.6/–5.7dB and +4.9/–4.8dB for the two frequency ranges is chosen to depict the upper and lower bounds of uncertainty in Figs. 3 and 4. The measured field strengths of the 12 sites agree within the maximum expanded uncertainty except at several high frequencies, as presented in Figs. 3 and 4.

5 Conclusions

An interlaboratory radiated emission comparison has been performed, comparing electric field strength measurements at 10m distance in the frequency range 30–1000MHz for 12 OATSs. A spherical dipole radiator is employed as a reference emitter. Comparison results are presented in terms of measurement repeatability of electric-field strengths measured at test sites and subsequent repeatability is 2.1 and 3.2dB (2σ) for horizontal and vertical polarisations, respectively.

Expanded uncertainty for the 95.45% confidence level is evaluated to be +5.6/–5.7dB and +4.9/–4.8dB for the frequency ranges corresponding to biconical and log-periodic antennas, respectively. Uncertainty analysis demonstrates that the uncertainty mainly stems from the antenna factor calibration, mismatch, the performance of a receiver and site imperfections.

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